

An Internship Report

Of

Google AI-ML

Submitted

To

**School of Computer Science and Engineering
IILM University, Greater Noida, U.P.**

In partial fulfilment of the requirement of degree of

B.Tech (CSE)



By

Roll Number of Student:- 2410031035

Name of Student: ANANT RAJ

Batch: 2024-28

CANDIDATE'S DECLARATION

I, ANANT RAJ, do hereby declare that the internship report titled “**Google AI-ML**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-

Student Name: ANANT RAJ

Roll No.: 2410031035

ACKNOWLEDGEMENT

I would like to take this opportunity to thank everyone who helped me through this internship. Their guidance, honest feedback, and general willingness to answer my questions made a genuine difference to how much I got out of the eight weeks.

I would like to thank the **AICTE – EduSkills Virtual Internship Program** team and **EduSkills Academy** for providing me the opportunity and the structured platform to learn and gain hands-on exposure in the domain of **Google AI-ML**, delivered in association with **Google for Developers** under the India Edu Program. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature:

Student Name: ANANT RAJ

Roll No.: 2410031035

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3. INTERNSHIP COMPLETION CERTIFICATE



Figure 3.1: Internship Completion Certificate – Google AI-ML Virtual Internship

INTERNSHIP OFFER LETTER



Ref No: C17Y2026M061447012

Internship Offer Letter

Student Name : Anant Raj
Stream : B.Tech
Branch : B.Tech(Computer Science and Engineering)
Institute Name : IILM University, Greater Noida
Internship Domain : Google AI-ML
Internship Duration : June 2026 - August 2026
AICTE Student ID : STU69986302ad9bd1771594498

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

Figure 3.2: Internship Offer Letter – Page 1

The week wise Structure of the Internship is as Follows:

Week	Module	Description
week 1	Program neural networks with TensorFlow	Everything you need to know to demystify machine learning, from first principles to creating convolutional neural networks for advanced image recognition and classification.
week 2	Get started with object detection	Basics of object detection and integrating pretrained object detectors into mobile apps.
week 3	Go further with object detection	Train custom object-detection models using TensorFlow Lite and Model Maker.
week 4	Get started with product image search	Build a product image search feature using on-device object detection.
week 5	Get started with product image search	Detect objects in static images and live camera feeds for visual product search.
week 6	Go further with product image search	Build backend integration for product image search mobile applications.
week 7	Go further with image classification	Build custom image-classification models and improve classification skills.
week 8	Go further with image classification	Create and integrate a custom image classifier model into your app.
Final Credential Validation / Internship Grade Point Assessment		

Note: There is no stipend associated with this internship.

We congratulate you on your selection and wish you a successful learning journey.


Bibek Ranjan
 Director, EduSkills

Scan to Verify



Supported by


 for Developers

Under the Patronage of



Figure 3.3: Internship Offer Letter – Page 2

4. PROJECT DESCRIPTION

4.1 Introduction

So, short version is: I did a Virtual Internship with Google AI-ML, under the AICTE – EduSkills Virtual Internship Program, for eight weeks, from June to August 2026, along with my regular B.Tech (Computer Science and Engineering) coursework at IILM University, Greater Noida. It was not a standard classroom module. It's a partner of Google for Developers under something called the India Edu Program and almost all of it is based on Google's public learning paths at developers.google.com. To be honest, I picked this program mostly because I got bored of reading about machine learning. Each week had a short video, codelab that I actually had to sit down and do, and a quiz that determined if I got that week's badge or not. By week eight I'd built a few models, got them working in mobile style workflows and picked up a few Google Developer badges along the way. At the very end there was a Final Credential Validation. I came out with a "O" (Outstanding) on the completion certificate EduSkills issued.

4.2 Organization Profile

Who is really driving this, just quickly. It is a project of EduSkills Academy, an ISO 9001:2015 and ISO 20000-1:2018 certified company working under the umbrella of "Nation Building Through Skills". They are working in collaboration with AICTE, Ministry of Education, Government of India, to operate the AICTE National Internship Portal.

Note that EduSkills is not actually creating its own lesson content for the Google AI-ML domain. It builds on top of Google's existing public learning pathways and codelabs, and Google's own Developer Profile badge system. I liked that part a lot – my profile badges aren't some EduSkills-only credential that no one outside the program recognizes. These are real Google Developer credentials and anyone can go to my public profile page and check them out.

4.3 Problem Statement

I believe this is a common problem among many engineering students. Myself among them. In class we learn the math of machine learning. Gradients, loss functions, backpropagation, the whole kit and caboodle. And then, often, nothing. It's still

speculation. The teaching includes topics such as convolutional neural networks, object detection and image classification, but not necessarily as a complete pipeline. No one really teaches you how a trained model becomes something that you can give to someone and say, “here, open this on your phone.”

This internship is an attempt to fill that gap or so. Once you have a trained model, it doesn’t end, it persists. Reduce it down using TensorFlow Lite, wire it into an app via ML Kit, couple it with a Cloud Vision API backend if the use case requires things like product search. At the end you have walked the entire path, not just the first half.

4.4 Project Objectives

If I had to crystallize what I really wanted from these eight weeks it looked something like this, less a list I made up beforehand, more something I built as I went along:

- Learn the ins and outs of neural networks in TensorFlow, from the first “Hello World” to convolutional networks for recognizing images.
- Discover how to integrate a pre-trained object detector into a mobile app using the Google ML Kit.
- Instead of using pre-trained object detection models, train your own using TensorFlow Lite and the Model Maker package.
- Built on-device product image search capability, starting with object identification.
- Learned to identify items in still photos and a live video stream - in practice, they behave quite differently, more so than I thought.
- Connect mobile frontend to real backend for searching product images
- Further pushing the image-classification models, and understanding what actually improves their accuracy.
- Integrate a custom image classifier into a functioning app, rather than keeping it as a notebook experiment.
- Have a public Google Developer Profile so the badges are visible and verifiable, and not just sitting in some EduSkills dashboard.

That last one especially - didn't think I'd care about it coming in but I did at the end .

4.5 Scope of the Project

On scope, I followed Google's approved Machine Learning pathway on developers.google.com. That is, six paths: "Program neural networks using TensorFlow", "Get started with object detection", "Go further with object detection", "Get started with product image search", "Go further with product image search" and "Go further with image classification". Same pattern every time - movie, codelab, quiz (at least 3 out of 4 needed to pass), badge.

In the last week I also created a public Google Developer Profile at a special g.dev/ address so everything I had earned lived in one location. I should be honest about the limitations here though, all of this was done in Google's educational codelab environment. None of it touched real production app or real Google Cloud infrastructure. I don't want to oversell it, it's a learning road not a deployment narrative.

4.6 Technologies and Tools Used

Not everything on this list is equally important - some of these I used a lot, some I used for one codelab and then never again. And so here's what I kind of worked with:

- TensorFlow – the main machine learning framework we use throughout.
- TensorFlow Lite and the Model Maker library for on-device model compression and preparation, yes indeed.
- Google ML Kit Object Detection API, for integrating trained models into a mobile app.
- Build backend for product image search feature using Google Cloud Vision API (Product Search)
- Python, mostly in Google Colab and browser-based codelabs.
- Google Developer Profile and Badges - credentialing mechanism for each pathway.
- * EduSkills Assess2ME – the portal for weekly quizzes and final examination. * The full curriculum is available at developers.google.com.

That's about eight different tools and platforms for an eight week curriculum which in hindsight feels about perfect.

4.7 System Architecture

Zoom out far enough and nearly all trails followed the same basic shape. Here's the thing.

- Model layer – you design and train a network in TensorFlow. Starting from the “Hello World of Machine Learning” and going up to CNNs for image tasks.
- Optimization layer — TensorFlow Lite and Model Maker compress that trained model to work on a phone rather than needing a server somewhere else.
- On-device inference — Google ML Kit takes the optimized model and performs real-time inference on an Android device.
- Backend layer—The Cloud Vision API analyzes the photos you take with a product library, not all on the device, especially for product search.
- Application layer – this is what the end user sees. The app will call the local model or the backend and will display whatever it gets back.
- Credentialing layer - Pass the quiz for a pathway and the badge appears on your public Google Developer profile.
- Assessment layer – Weekly Assess2ME quizzes to check whether the topics really stuck and it all ends up with a Final Credential Validation.

4.8 Methodology

It was pretty much the same week after week, a little concept learning, a hands-on codelab, a knowledge-check quiz, and a badge waiting for me at the end if I passed it. That's how it actually broke down week by week.

Table 4.1: Week-wise Structure of the Google AI-ML Virtual Internship

Week	Module	Description
1	Program neural networks with TensorFlow	Everything you need to know to demystify machine learning, from first principles to creating convolutional neural networks for advanced image recognition and classification.
2	Get started with object detection	Basics of object detection and integrating pretrained object detectors into mobile apps.
3	Go further with object detection	Train custom object-detection models using TensorFlow Lite and Model Maker.
4	Get started with product image search	Build a product image search feature using on-device object detection.
5	Get started with product image search	Detect objects in static images and live camera feeds for visual product search.
6	Go further with product image search	Build backend integration for product image search mobile applications.

Week	Module	Description
7	Go further with image classification	Build custom image-classification models and improve classification skills.
8	Go further with image classification	Create and integrate a custom image classifier model into your app.
Final Credential Validation / Internship Grade Point Assessment		

At the end of each week there was a graded quiz on the EduSkills Assess2ME webpage – I had to get at least 3 out of 4 to pass. Cleared it and the corresponding Google Developer badge unlocked: The TensorFlow badge “Get started with object detection” “Go further with object detection” “Start with product image search” “Go further with product image search” and finally “Go further with image classification” Each one was taken to a public Google Developer Profile I created with a custom web address (g.dev/). All that mattered was the final Credential Validation and the score I received was a combination of my performance on all the weekly quizzes and that final exam.

4.9 Expected Outcomes

So what did I walk away with eight weeks later? A few things mostly: • Practical knowledge of how to make and train a simple neural network using TensorFlow, not just theoretically, but practically starting from the “Hello World of Machine Learning” example.

- Good knowledge of wiring a pre-trained object detector into a mobile app using Google.MLKIT
- Hands-on experience training and deploying a custom object detection model utilizing TensorFlow Lite and Model Maker
- Ability to build a product image search function using on-device detection with a Cloud Vision API backend.
- At ease with image classification to create, train and integrate a custom classifier into a working app.
- A public verified Google Developer Profile with all the badges I earned along the way.
- More confidence going into practical, on-device AI/ML work than I had going in, to be honest.

4.10 Certificates of Completion and Communication Proof

Chapter 3 of this report has both the Internship Offer Letter and the Internship Completion Certificate from EduSkills Academy (Figures 3.1–3.3) – together they confirm I finished the 8-week AICTE – EduSkills Virtual Internship in Google AI-ML with an “O” (Outstanding) grade. Figure 4.1 below shows a few of the Google Developer badges I picked up along the way, just as extra proof that each module actually got completed.



Figure 4.1: Google Developer Badges Earned During the Internship

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